Problems: Bootstrap CLT Normal

Monday, October 19, 2026 3:37 PM

1. $\bar{X} = 52, \quad s = 10, \quad n = 100$

Standard error: $\frac{10}{\sqrt{100}} = 1$

Z-score:

$$\frac{52 - 50}{10/\sqrt{100}} = \frac{2}{1} = 2$$

- At 90%: $2 > 1.645$ REJECT NULL
- At 95%: $2 > 1.96$ REJECT NULL
- At 99%: $2 < 2.576$ FAIL TO reject Null

95% Confidence Interval:

$$52 \pm 1.96(1) = [50.04, 53.96]$$

2. $\hat{S} = 3.1 \quad \widehat{SE}(\hat{S}) = 1.3$

90%: $3.1 \pm \underbrace{1.3(1.645)}_{2.1385} = [0.9615, 5.2385]$

- 0 not in CI, REJECT NULL

95%: $3.1 \pm \underbrace{1.3(1.96)}_{2.548} = [0.552, 5.648]$

- 0 not in CI, REJECT NULL

99%: $3.1 \pm \underbrace{1.3(2.576)}_{3.3488} = [-0.2488, 6.4488]$

- 0 in CI, FAIL TO Reject Null

- We fail to reject the null hypothesis when the test value falls in the confidence interval

3. $\bar{X} = 11.5, \quad s = 3, \quad n = 36$

$H_0: \mu = 12 \quad H_1: \mu \neq 12$

$SE(\bar{X}): \quad \frac{3}{\sqrt{36}} = 0.5$

95% CI: $11.5 \pm 0.5(1.96) = [10.52, 12.48]$

- 12 falls in the confidence interval, fail to reject null at 95% confitence.

4. Group A: $\bar{X} = 25, \quad s = 8, \quad n = 100$

Group B: $\bar{X} = 22, \quad s = 7, \quad n = 100$

$$Z = \frac{25 - 22}{\sqrt{\frac{8^2}{100} + \frac{7^2}{100}}} = \frac{3}{\sqrt{\frac{64}{100} + \frac{49}{100}}} = \frac{3}{\underset{\overset{\nearrow}{SE}}{1.063}} = 2.82$$

95% Confidence Interval:

$$\underbrace{(25 - 22)}_{3} \pm \underbrace{1.063(1.96)}_{} = [0.917, 5.083]$$

$$3 \pm 2.083$$

5. A: $\bar{X} = 14, \quad s = 6, \quad n = 81$

B: $\bar{X} = 12, \quad s = 5, \quad n = 64$

$$SE = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \sqrt{\frac{36}{81} + \frac{25}{64}}$$

$$= 0.914$$

test statistic $= \frac{14 - 12}{0.914} = 2.188$

6. A: $\bar{X} = 3.2, \quad s = 1.5, \quad n = 49$

B: $\bar{X} = 2.8, \quad s = 1.4, \quad n = 64$

Test statistic: $\frac{3.2 - 2.8}{\sqrt{\frac{1.5}{49} + \frac{1.4}{64}}} = \frac{0.4}{\sqrt{0.07654}}$

$= \frac{0.4}{0.2767}$

$= 1.445$

$1.445 < 1.96$

Not Statisticclly-Significant at 95%

CI: $0.4 \pm \underbrace{0.2767(1.96)}_{}$

$[-0.142, 0.942]$

0 not in CI, fail to reject the null.

7-10: See .ipynb File